DBMS Fundamentals — Test Notes

1. Database Management System (DBMS)

A Database Management System is software used to create, store, organize, retrieve, and manage data in databases. It helps maintain data security, consistency, integrity, and controlled access.

2. Advantages of DBMS

• Reduces data redundancy.
• Improves data consistency.
• Provides security and access control.
• Supports backup and recovery.
• Allows concurrent access by multiple users.

3. Relational Database

A relational database stores data in tables made up of rows and columns. Keys are used to identify records and establish relationships between tables.

4. Important Keys

A primary key uniquely identifies each row in a table. A foreign key refers to a key in another table and helps establish relationships.

5. SQL

SQL (Structured Query Language) is used to define, manipulate, and query relational data. Common commands include SELECT, INSERT, UPDATE, DELETE, CREATE, and ALTER.

6. Normalization

Normalization organizes data to reduce redundancy and avoid insertion, update, and deletion anomalies. Common normal forms include 1NF, 2NF, and 3NF.

7. Transactions and ACID

A transaction is a logical unit of database work. ACID stands for Atomicity, Consistency, Isolation, and Durability.

Sample Questions for LearnMate AI

Question	Expected topic
What is a DBMS?	DBMS definition
What are the advantages of DBMS?	Advantages
What is a primary key?	Keys
What is normalization?	Normalization
What does ACID stand for?	Transactions